## **Question 1:** SJF Non-Preemptive Scheduling Calculation.

| **Process** | **AT** | **BT** |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 8 |
| P3 | 2 | 7 |
| P4 | 3 | 3 |

1.Draw the Gantt chart for Shortest Job First (**Non-Preemptive**) scheduling.

| **Process** | **P1** | **P4** | **P3** | **P2** |
|---|---|---|---|---|
| **Time** | 0 – 6 | 6 - 9 | 9 - 16 | 16 - 24 |

2.Calculate Completion Time (**CT**), Turnaround Time (**TAT**), and Waiting Time (**WT**) for each process.

| **Process** | **AT** | **BT** | **CT** | **TAT(CT-AT)** | **WT(TAT-BT)** |
|---|---|---|---|---|---|
| **P1** | 0 | 6 | 6 | 6 | 0 |
| **P2** | 1 | 8 | 24 | 23 | 15 |
| **P3** | 2 | 7 | 16 | 14 | 7 |
| **P4** | 3 | 3 | 9 | 6 | 3 |

3.Compute the Average Waiting Time (**AWT**) and Average Turnaround Time (**ATAT**).

Here,

$$\text{Average Waiting Time (AvgWT)} = \frac{6+23+14+16}{4} = 12.25$$

## **Question 2**: SJF Non-Preemptive C Program Implementation.

Write a C program to implement Shortest Job First (SJF) Non-Preemptive CPU Scheduling.

The program should:

1. Take number of processes, arrival times, and burst times as input.

2. Calculate WT, TAT, CT for each process.

3. Print Average WT and Average TAT.

4. Show a simple textual Gantt chart.

## Code:

```
#include <bits/stdc++.h>
#include <iomanip>
using namespace std;

int main()
{
    int n;
    cout << "Enter number of processes: ";
    cin >> n;

    int at[100], bt[100], tat[100], wt[100];
    int completed[100];
    int gantt[100];
    int time_mark[101];

    for(int i=0;i<n;i++)
        completed[i] = 0;

    cout << "Enter Arrival Time and Burst Time\n";

    for(int i=0;i<n;i++)
    {
        cout << "P" << i+1 << ": ";
        cin >> at[i] >> bt[i];
    }
```

```
int current_time = 0;
int completed_count = 0;
int k = 0;

time_mark[0] = 0;

while(completed_count < n)
{
    int idx = -1;
    int min_bt = 9999;

    for(int i=0;i<n;i++)
    {
        if(at[i] <= current_time && completed[i] == 0)
        {
            if(bt[i] < min_bt)
            {
                min_bt = bt[i];
                idx = i;
            }
        }
    }

    if(idx != -1)
    {
        current_time += bt[idx];

        tat[idx] = current_time - at[idx];
```

```
        wt[idx] = tat[idx] - bt[idx];

        completed[idx] = 1;

        gantt[k] = idx;
        time_mark[k+1] = current_time;

        k++;
        completed_count++;
    }
    else
    {
        current_time++;
    }
}

float avgWT = 0, avgTAT = 0;

cout << "\n";
cout << left << setw(10) << "Process"
     << setw(10) << "AT"
     << setw(10) << "BT"
     << setw(10) << "WT"
     << setw(10) << "TAT" << endl;

for(int i=0;i<n;i++)
{
    cout << left << setw(10) << ("P"+to_string(i+1))
         << setw(10) << at[i]
```

```
        << setw(10) << bt[i]
        << setw(10) << wt[i]
        << setw(10) << tat[i] << endl;

      avgWT += wt[i];
      avgTAT += tat[i];
   }

   cout << "\nAverage WT = " << avgWT/n;
   cout << "\nAverage TAT = " << avgTAT/n;

   cout << "\n\nGantt Chart:\n";

   for(int i=0;i<k;i++)
      cout << "| P" << gantt[i]+1 << " ";

   cout << "|\n";

   for(int i=0;i<=k;i++)
      cout << time_mark[i] << " ";

   cout << endl;

   return 0;
}
```

**OUTPUT:**

"E:\7th semester\OS LAB\jobschedulingAlog.exe"

```
Enter number of processes: 4
Enter Arrival Time and Burst Time
P1: 0 6
P2: 1 8
P3: 2 7
P4: 3 3

Process   AT        BT        WT        TAT
P1        0         6         0         6
P2        1         8         15        23
P3        2         7         7         14
P4        3         3         3         6

Average WT = 6.25
Average TAT = 12.25

Gantt Chart:
| P1 | P4 | P3 | P2 |
0 6 9 16 24

Process returned 0 (0x0)   execution time : 35.506 s
Press any key to continue.
```

## **Question 3:** Analytical Comparison

Compare **SJF** Non-Preemptive with **FCFS** for the following processes:

| Process | AT | BT |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 8 |
| P3 | 2 | 7 |
| P4 | 3 | 3 |

**(i)** By using SJF:

Average Waiting Time (AvgWT)= $\frac{0+15+7+3}{4} = 6.25$

Average Turnaround Time (AvgTAT)= $\frac{6+23+14+16}{4} = 12.25$

**(ii)** By using FCFS:

| **Process** | **P1** | **P4** | **P3** | **P2** |
|---|---|---|---|---|
| **Time** | 0 – 6 | 6 - 14 | 14 - 21 | 21 - 24 |

So,

| Process | AT | BT | CT | TAT(CT-AT) | WT(TAT-BT) |
|---|---|---|---|---|---|
| **P1** | 0 | 6 | 6 | 6 | 0 |
| **P2** | 1 | 8 | 14 | 13 | 5 |
| **P3** | 2 | 7 | 21 | 19 | 12 |
| **P4** | 3 | 3 | 24 | 21 | 18 |

Average Waiting Time (AvgWT)= $\frac{0+5+12+18}{4} = 8.75$

Average Turnaround Time (AvgTAT)= $\frac{6+13+19+21}{4} = 14.75$

**In conclusion, Average Waiting Time and the Average Turnaround Time in SJF schedule is less than the FCFS schedule process. Thus, the SJF schedule process is better than FCFS**

schedule process for this given process.